



RAMCO INSTITUTE OF TECHNOLOGY

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Department of Information Technology

Academic Year 2023 – 2024 (Even Semester)

Degree, Semester & Branch: IV Semester B.Tech. IT

Course Code & Title: CS3491 & Artificial Intelligence and Machine Learning

Name of the Faculty member (s): Mrs.G.Mareeswari

Innovative Practice Description

Unit / Topic: Unit 5 / Hyper Parameter Tuning

Course Outcome: CO5

Activity Chosen: Think Pair Share

Justification:

- Think pair share activity involves the students to think individually and to share the knowledge.
- Hyper parameter tuning is an essential aspect of machine learning process. Hyperparameter tuning is the process of selecting the optimal values for a machine learning model's hyperparameters. A good choice of hyperparameters can really make a model succeed in meeting desired metric value or on the contrary it can lead to an unending cycle of continuous training and optimization. The students' individual understanding capability can be tested and the gained ideas can be shared by think-pair-share activity.

Time Allotted for the Activity: 15 minutes

Details of the Implementation:

- Instructor explained the particular concepts/topic in classroom within 30 minutes. Instructor gave the list of questions to the students.
- Students think about what they know or have learned in the Hyper parameter tuning concept, and come up with their own idea. [Takes 1-5 Minutes].
- The instructor asks them to pair with nearby students to discuss the answers. The students interact with each other about their answers to deepen understanding of the topic.
- They share their thoughts with each other and proceed with the task. [Takes 6- 10 Minutes].
- Students share their answers with the entire class [Takes 11-15 Minutes].
Instructor moderates the discussion and highlights important points.

CO – PO / PSO mapping:

CO	PO1	PO2	PO3	PO9	PO10	PSO1
CO 3	2	1	1	1	1	1

(1 – Low 2 – Moderate 3 – High)

PO / PSO mapped:

Innovative practice	PO1	PO2	PO3	PO9	PO10	PSO1
	2	1	1	1	1	1
Justification for correlation	It includes logical thinking and make the student able to think about problem solving approach	Able to approach the program by applying what they learnt.	Able to provide solution for the machine learning related programs.	Students involved in the activity as team, the team work Skill Improved.	Students' communication skill will be improved as they are facing audience and communicating the activity	Students were able to apply hyper parameter tuning in the field of data science to provide solution for the complex computational problems.

• Images / Screenshot of the practice:

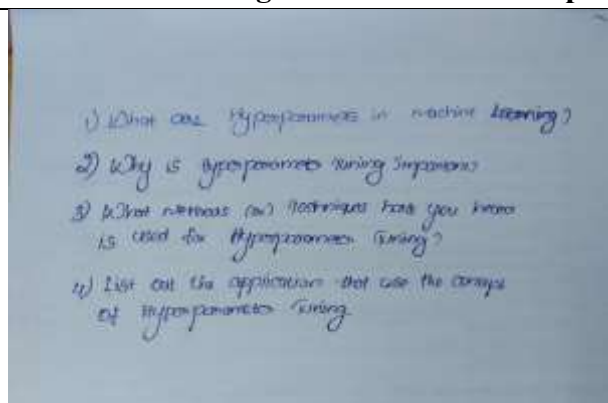


Figure. 1a



Figure. 1b



Figure. 1c



Figure. 1d

Glimpse of Think Pair Share Activity

(Figure.1a shows the list of questions shared to the students, Figure1.b shows team discussion of Aarthi.S, Abinaya. S. G, and Yazhini, Figure.1c shows discussion of Sivasekar and Praveen Kumar, Figure.1d shows discussion of Sneha and Sandhiya with the class members)

❖ **Feedback of practice from students and other stakeholders:**

- The students felt that this activity helps them to remember the concepts and it will be very helpful to correct their understanding about the topic while sharing the answers to the other students.
- The students felt that this activity is so interesting and helps them to remember
- They said it will be very helpful when preparing for exams

❖ **Benefit of the practice:**

- From this activity, the students can get more clarity in the hyper parameter tuning topic.
- Students can able to explain the concepts without any confusion.
- This practice has made the class more interactive than the conventional practice

❖ **Challenges faced in implementation:**

- All the students were unable to complete the task within allotted time
- Few students found it difficult to solve the recollect the concept individually. After discussion, they felt easy

References:

1. <https://www.ritrjpm.ac.in/images/computer-science/TPS-Venkat-GE6075.pdf>
2. <https://www.readingrockets.org/strategies/think-pair-share>
3. <https://www.wgu.edu/heyteach/article/how-think-pair-share-activity-can-improve-your-classroom-discussions1704.html>
4. https://www.ritrjpm.ac.in/images/B.Tech-IT/20232024/IP/GM_GE3151_U3_Think_pair.pdf

Signature of Faculty Member

HOD